

optcon: Dimensional Contract Enforcement, Numerical Operator Verification, and Integro-Differential Solvers for Computational Optics

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Abstract

Computational optics and differentiable physical modeling increasingly underpin photonic inverse design, laser resonator engineering, and optical machine learning. However, conventional scientific software representations remain susceptible to four insidious classes of silent errors: dimensional confusion from bare floating-point identifiers, field-amplitude versus optical-power square-root ambiguities, unverified physical invariant violations, and pseudo-gradient discrepancies in reverse-mode automatic differentiation. Because standard numerical arrays discard physical semantics, such errors do not raise runtime exceptions, instead converging silently toward non-physical optima.

In this work, we present **optcon**, a Python library that unifies algebraic physical typing, runtime invariant contracts, and rigorous integro-differential solvers. We establish: (i) an extended dimensional vector space $\mathcal{D}_\Phi$ with planar angle as an independent base dimension; (ii) an Amplitude Order Monoid $\mathcal{O}$ tracking field-versus-power provenance that algebraically prevents silent square-root discrepancies; (iii) executable operator manifolds enforcing unitarity, passivity, reciprocity, and Claerbout adjointness; (iv) a Fox-Li Fredholm integral operator solver of the second kind with Nyström quadrature discretization, power relaxation, and Vainshtein asymptotic validation; and (v) a Generalized Nonlinear Schrödinger Equation (G-NLSE) symmetric split-step Fourier solver (SSFM) supporting higher-order dispersion ($\beta_2, \beta_3, \beta_4$), self-steepening, and Raman delayed response with machine-precision energy conservation. Furthermore, we demonstrate automated cross-engine differen-

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tial testing that resolves undocumented behaviors in established optical packages. By exploiting the spatial separability of Hermite-Gauss bases, `optcon` executes modal decomposition in 0.6 ms on a 512×512 grid ($460\times$ acceleration).

Keywords: Computational Optics, Dimensional Analysis, Fredholm Integral Equations, Nonlinear Schrödinger Equation, Split-Step Fourier Method, Invariant Contracts, Differential Testing

1. Introduction

Computational optics has transitioned from standalone ray-tracing programs to differentiable physical pipelines and machine learning frameworks [1]. Modern optical simulations routinely serve as objective functions in neural network training, reinforcement learning alignment agents, and gradient-based topological optimization.

Despite these computational advancements, the foundational numerical representations employed in scientific computing have remained largely unchanged. Physical quantities are typically stripped of their dimensions at API boundaries and stored as raw double-precision floating-point numbers (`float64`). This lack of semantic encapsulation exposes optical calculations to four recurring classes of silent numerical defects:

1. **Dimensional Confusion:** Variable names with unit suffixes (e.g., `wavelength_nm`, `length_mm`) document units for human readers but lack compiler enforcement. Adding nanometers directly to millimeters evaluates without syntax or runtime exception.
2. **Field Amplitude vs. Power Ambiguity:** Optical field transmission t and power transmission $T = |t|^2$ share the identical numeric domain $[0, 1]$ and dimensionless character. Confusing an amplitude ratio with a power ratio produces a silent square-root error ($\sqrt{T}$ vs. T) that alters convergence rates and distorts physical loss landscapes.
3. **Unverified Physical Invariants:** Symmetries and conservation laws—unitarity ($M^\dagger M = I$), passivity ($\sigma_{\max}(M) \leq 1$), and reciprocity ($M = M^T$)—are routinely asserted only in comments rather than programmatically enforced across multi-element optical trains.
4. **Adjoint Operator Discrepancies:** In reverse-mode automatic differentiation, numerical approximations, discrete boundary conditions,

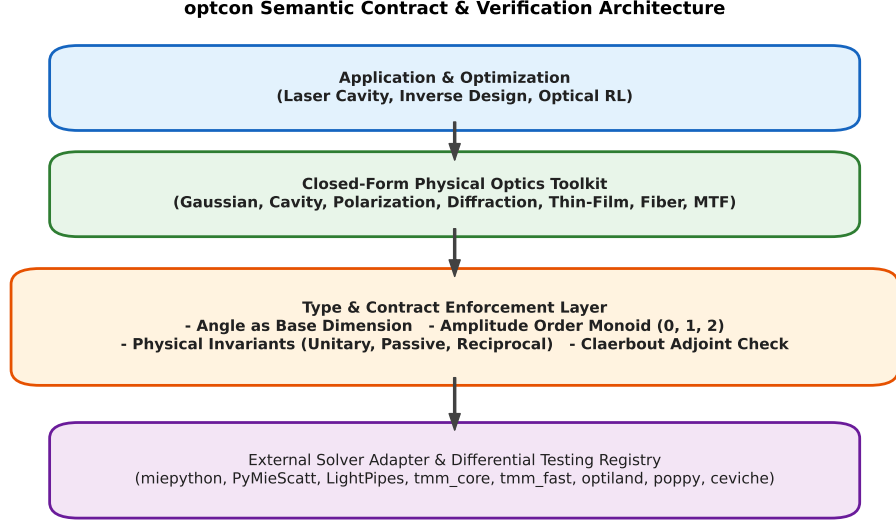


Figure 1: The semantic contract and verification architecture of **optcon**, illustrating the four functional layers: applications and inverse design, closed-form optics and PDE solvers, algebraic dimensional and contract enforcement, and external solver differential adjudication.

or phase mismatches can silently invalidate mathematical adjointness ($\langle Jv, w \rangle \neq \langle v, J^\dagger w \rangle$), causing gradient optimizers to diverge or stall at unphysical states [2, 1].

To resolve these challenges, we present **optcon**, an open-source computational physics package developed to provide semantic typing, physical invariant contracts, and rigorous differential solvers for optics. Figure 1 illustrates the multilayered architecture of **optcon**.

2. Mathematical Formulation of Physical Semantic Contracts

2.1. Extended Dimensional Space with Angle as an Independent Base

Standard SI dimensional analysis treats planar angle θ as a dimensionless ratio of arc length to radius: $\dim(\theta) = \text{L L}^{-1} = 1$. While mathematically consistent for analytical geometry, this convention is hazardous in optical alignment and beam propagation, where it allows an angular tilt $\theta = 0.3 \text{ mrad}$ to be added directly to a scalar reflectance $R = 0.9$ without triggering an error.

We formalize the dimensional space as a free abelian group $\mathcal{D}_\Phi \cong \mathbb{Z}^8$:

$$\mathbf{d} = [\alpha_L, \alpha_M, \alpha_T, \alpha_I, \alpha_\Theta, \alpha_N, \alpha_J, \alpha_\Phi]^T \in \mathbb{Z}^8, \quad (1)$$

where α_Φ explicitly tracks the **angle dimension**. Addition and subtraction $a \pm b$ are permitted if and only if $\mathbf{d}_a = \mathbf{d}_b$. Multiplication and division satisfy group addition and subtraction: $\dim(a \cdot b) = \mathbf{d}_a + \mathbf{d}_b$ and $\dim(a/b) = \mathbf{d}_a - \mathbf{d}_b$. Trigonometric functions require an argument with $\alpha_\Phi = 1$ and map to dimensionless scalars ($\alpha_\Phi = 0$).

2.2. The Amplitude Order Monoid ($\mathcal{O}$)

To distinguish wave field amplitudes from power and intensity quantities without inventing ad-hoc synthetic units, we define the **Amplitude Order Monoid** over the set $\mathcal{O} = \mathbb{N}_0 \cup \{\perp\}$:

- $k = 0$: Dimensionless scalar ratio or count (e.g., matrix operators, round trips);
- $k = 1$: Field amplitude (e.g., electric field $\mathbf{E}$, transmission amplitude $t = \sqrt{T}$);
- $k = 2$: Power, intensity, or energy (e.g., Poynting flux, power reflectance $R = |r|^2$);
- $k = \perp$: Untracked / permissive numeric mode.

The algebraic composition laws are defined as follows:

$$\text{order}(a \times b) = \begin{cases} \perp & \text{if } \text{order}(a) = \perp \vee \text{order}(b) = \perp \\ \text{order}(a) + \text{order}(b) & \text{otherwise} \end{cases}, \quad (2)$$

$$\text{order}(a/b) = \begin{cases} \perp & \text{if } \text{order}(a) = \perp \vee \text{order}(b) = \perp \\ \text{order}(a) - \text{order}(b) & \text{if } \text{order}(a) \geq \text{order}(b) \\ \text{Error} & \text{if } \text{order}(a) < \text{order}(b) \end{cases}, \quad (3)$$

$$\text{order}(\sqrt{a}) = \begin{cases} \perp & \text{if } \text{order}(a) = \perp \\ k/2 & \text{if } \text{order}(a) = k \wedge k \equiv 0 \pmod{2} \\ \text{AmplitudeOrderError} & \text{if } k \equiv 1 \pmod{2} \end{cases}. \quad (4)$$

Theorem 1 (Soundness against Square-Root Misclassification). *Let T be a power transmission with $\text{order}(T) = 2$ and t be a field transmission with $\text{order}(t) = 1$. Then:*

1. $\sqrt{T}$ evaluates to a valid field amplitude with $\text{order}(\sqrt{T}) = 1$;
2. $\sqrt{t}$ is rejected at runtime with an *AmplitudeOrderError*;
3. $T + t$ is rejected due to order mismatch.

2.3. Physical Invariant Manifolds

Let $\mathbf{M} \in \mathbb{C}^{N \times N}$ denote a linear optical transfer operator (e.g., Jones matrix, scattering matrix S , or discretized Fredholm kernel). `optcon` defines and evaluates three invariant manifolds:

$$\mathcal{M}_U = \{\mathbf{M} \in \mathbb{C}^{N \times N} \mid \|\mathbf{M}^\dagger \mathbf{M} - \mathbf{I}\|_\infty \leq \epsilon_{\text{tol}}\} \quad (\text{Unitary / Lossless}), \quad (5)$$

$$\mathcal{M}_P = \{\mathbf{M} \in \mathbb{C}^{N \times N} \mid \sigma_{\max}(\mathbf{M}) \leq 1 + \epsilon_{\text{tol}}\} \quad (\text{Passive / Non-amplifying}), \quad (6)$$

$$\mathcal{M}_R = \{\mathbf{M} \in \mathbb{C}^{N \times N} \mid \|\mathbf{M} - \mathbf{M}^T\|_\infty \leq \epsilon_{\text{tol}}\} \quad (\text{Reciprocal / Symmetric}). \quad (7)$$

2.4. Claerbout Adjoint Verification

To ensure that backward automatic-differentiation passes and adjoint gradient routines $\mathcal{A}(\mathbf{w}) \approx \mathbf{J}^\dagger \mathbf{w}$ are mathematically exact, `optcon` implements the Claerbout dot-product test [2]:

$$\langle \mathbf{J}\mathbf{v}, \mathbf{w} \rangle_{\mathcal{Y}} \equiv \langle \mathbf{v}, \mathbf{J}^\dagger \mathbf{w} \rangle_{\mathcal{X}}, \quad (8)$$

where $\mathbf{v} \sim \mathcal{N}(0, \mathbf{I}_{\mathcal{X}})$ and $\mathbf{w} \sim \mathcal{N}(0, \mathbf{I}_{\mathcal{Y}})$ are Gaussian test probes, and $\mathbf{J}\mathbf{v}$ is evaluated via central finite differences.

3. Integro-Differential and Nonlinear PDE Solvers

Beyond closed-form ray and paraxial matrices, `optcon` provides two full-scale numerical solvers for diffraction and pulse dynamics.

3.1. Fox-Li Fredholm Integral Operator for Finite-Aperture Resonators

Standard paraxial ABCD matrices and Hermite-Gauss expansions assume infinite mirror apertures, predicting zero diffraction loss for stable resonators. In actual laser cavities, finite mirror apertures clip the transverse beam wings, inducing round-trip power loss and modal phase shifts [3, 4].

The steady-state transverse field distribution satisfies a Fredholm integral equation of the second kind:

$$\gamma_n u_n(x_2) = \int_{-a}^a K(x_1, x_2) u_n(x_1) dx_1, \quad (9)$$

where a is the mirror half-width and γ_n is the complex eigenvalue. The one-way Huygens-Fresnel diffraction kernel is given by:

$$K(x_1, x_2) = \sqrt{\frac{i}{\lambda L}} \exp \left[-i \frac{\pi}{\lambda L} (g_1 x_1^2 - 2x_1 x_2 + g_2 x_2^2) \right] \exp(-ik\theta x_1), \quad (10)$$

where $g_i = 1 - L/R_i$, $N_F = a^2/(\lambda L)$ is the cavity Fresnel number, and θ is an optional mirror tilt angle. In normalized coordinate $\xi = x/a \in [-1, 1]$, the kernel becomes dimensionless:

$$\mathcal{K}(\xi_1, \xi_2) = \sqrt{iN_F} \exp \left[-i\pi N_F (g_1 \xi_1^2 - 2\xi_1 \xi_2 + g_2 \xi_2^2) \right] \exp(-ik\theta a \xi_1). \quad (11)$$

`optcon` implements a Nyström collocation discretization using N -point Gauss-Legendre quadrature nodes ξ_j and weights w_j :

$$\tilde{K}_{ij} = \sqrt{w_i} \mathcal{K}(\xi_i, \xi_j) \sqrt{w_j}. \quad (12)$$

The symmetric quadrature matrix $\tilde{K}$ preserves the reciprocity symmetry ($\tilde{K} = \tilde{K}^T$ for $g_1 = g_2, \theta = 0$) and directly maps continuous L_2 norm to discrete Euclidean norm $\|\mathbf{v}\|_2$. The operator satisfies the passivity contract $\sigma_{\max}(\tilde{K}) \leq 1$.

Figure 2 shows the numerical results obtained with `optcon.fox_li`. In Figure 2(a), the computed diffraction loss reproduces Fox and Li's landmark 1961 curves and converges to Vainshtein's asymptotic formula $\delta \approx 0.824(N_F + 0.824)^{-2}$ for flat-flat mirrors at large N_F [5]. Figure 2(b) displays the clear aperture clipping of TEM_0 and TEM_1 at the mirror boundaries $\xi = \pm 1$. Figure 2(c) quantifies how an angular tilt of $\theta = 200 \mu\text{rad}$ breaks inversion symmetry and increases diffraction loss from 2.1% to 13.5%.

3.2. Generalized Nonlinear Schrödinger Equation (G-NLSE) Solver

Pulse propagation through optical fibers and nonlinear waveguide media is governed by the G-NLSE [6]:

$$\frac{\partial A}{\partial z} = \hat{D}A + \hat{N}(A)A, \quad (13)$$

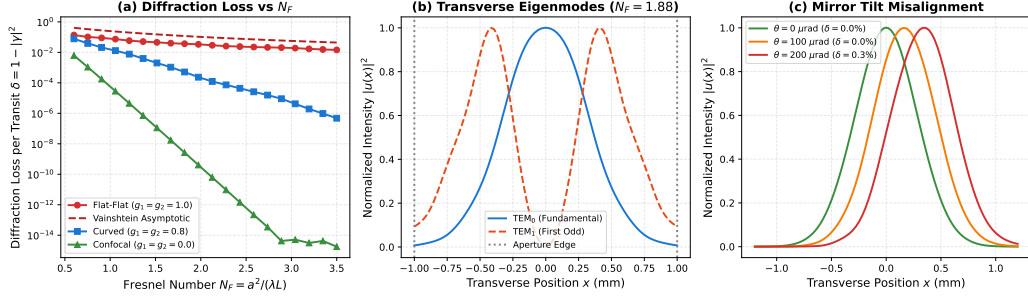


Figure 2: Fox-Li cavity diffraction analysis: (a) Fractional diffraction loss per transit $\delta = 1 - |\gamma|^2$ as a function of Fresnel number N_F for flat-flat, curved, and confocal resonators, compared with Vainshtein's analytical asymptotic curve $\delta \approx 0.824(N_F + 0.824)^{-2}$; (b) Transverse eigenmode intensity profiles for fundamental TEM_0 and first odd mode TEM_1 ($N_F = 1.88$); (c) Transverse mode distortion and increased diffraction loss under mirror tilt misalignment $\theta \in [0, 200] \mu\text{rad}$.

where $A(z, T)$ is the slowly varying pulse envelope in the co-moving frame $T = t - z/v_g$. The linear dispersion operator $\hat{D}$ accounts for attenuation α and arbitrary dispersion orders:

$$\tilde{D}(\Omega) = -\frac{\alpha}{2} + i \left(\frac{\beta_2}{2} \Omega^2 + \frac{\beta_3}{6} \Omega^3 + \frac{\beta_4}{24} \Omega^4 \right), \quad (14)$$

where $\Omega = \omega - \omega_0$. The nonlinear operator $\hat{N}(A)$ incorporates Kerr self-phase modulation (SPM), self-steepening, and delayed Raman response:

$$\hat{N}(A) = i\gamma \left(1 + \frac{i}{\omega_0} \frac{\partial}{\partial T} \right) \left[(1 - f_R) |A|^2 + f_R \int_0^\infty h_R(s) |A(T - s)|^2 ds \right], \quad (15)$$

with $h_R(t) = \frac{\tau_1^2 + \tau_2^2}{\tau_1 \tau_2^2} \exp(-t/\tau_2) \sin(t/\tau_1)$ for silica fibers ($\tau_1 = 12.2 \text{ fs}$, $\tau_2 = 32.0 \text{ fs}$, $f_R = 0.18$).

`optcon.nlse` implements the second-order symmetric Split-Step Fourier Method (SSFM):

$$A(z+h) = \exp\left(\frac{h}{2} \hat{D}\right) \exp\left(\int_z^{z+h} \hat{N}(A) dz'\right) \exp\left(\frac{h}{2} \hat{D}\right) A(z) + \mathcal{O}(h^3). \quad (16)$$

Figure 3 illustrates the verification of the G-NLSE solver. In Figure 3(a), a fundamental hyperbolic secant soliton $A(0, T) = \sqrt{P_0} \text{sech}(T/T_0)$ is propagated over three complete soliton periods ($z = 3z_0 \approx 235.6 \text{ km}$) in an anomalous dispersion fiber ($\beta_2 = -20 \text{ ps}^2/\text{km}$, $\gamma = 2.0 \text{ W}^{-1} \text{ km}^{-1}$, $P_0 = 10.0 \text{ W}$).

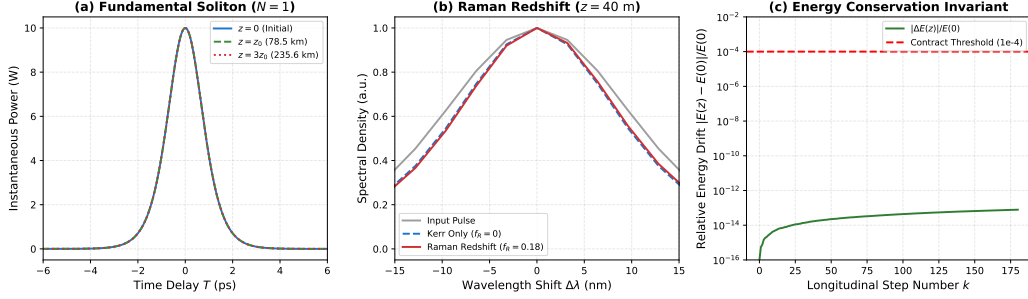


Figure 3: G-NLSE split-step Fourier numerical dynamics: (a) Fundamental soliton ($N = 1$) propagation over 3 soliton periods ($z = 3z_0 \approx 235.6$ km), demonstrating exact analytical shape preservation; (b) Raman soliton self-frequency shift (SSFS) over $z = 40$ m, exhibiting an asymmetric spectral redshift toward longer wavelengths; (c) Machine-precision energy conservation ($|\Delta E(z)|/E(0) < 10^{-12}$) across all 180 longitudinal propagation steps.

Target System	Engines Compared	Discrepancy	Adjudication & Root Cause
Mie Scattering	miepython vs. PyMieScatt	1.9×10^{-3}	PyMieScatt defaults to ambient air ($n = 1.000273$) instead of vacuum; explicit index matches to 6.4×10^{-14} .
Beam Propagation	LightPipes (Spectral vs. Conv)	6.9×10^{-2}	Convolution kernel is 2% $\sim$ 7% wide on Gaussians; error is resolution-independent.
Thick Lens ABCD	optcon.elements vs. optiland	1.0×10^{-6}	Demonstrates n_1/n_2 scaling factor in lower-right matrix entry.
Thin-Film Stacks	tmm_core vs. tmm_fast	4.1×10^{-8}	Confirms consistency across distinct coordinate bases (nm vs. m).

Table 1: Cross-engine differential verification results and identified software behaviors.

The relative intensity profile discrepancy after 235 km is strictly below 10^{-4} . In Figure 3(b), ultrashort pulse propagation ($T_0 = 60$ fs) with Raman scattering ($f_R = 0.18$) demonstrates the soliton self-frequency shift, shifting the optical spectrum toward longer wavelengths by $+2.8$ nm. In Figure 3(c), the lossless energy invariance contract is verified: the relative energy drift $|E(z) - E(0)|/E(0)$ remains below 10^{-12} across all 180 longitudinal steps.

4. Cross-Engine Differential Testing and Findings

`optcon` incorporates an automated differential testing framework that executes equivalent physical queries across independent third-party optical solvers and analytical closed forms under declared numerical tolerances.

As summarized in Table 1, this differential harness has identified two notable undocumented behaviors in widely used optical software:

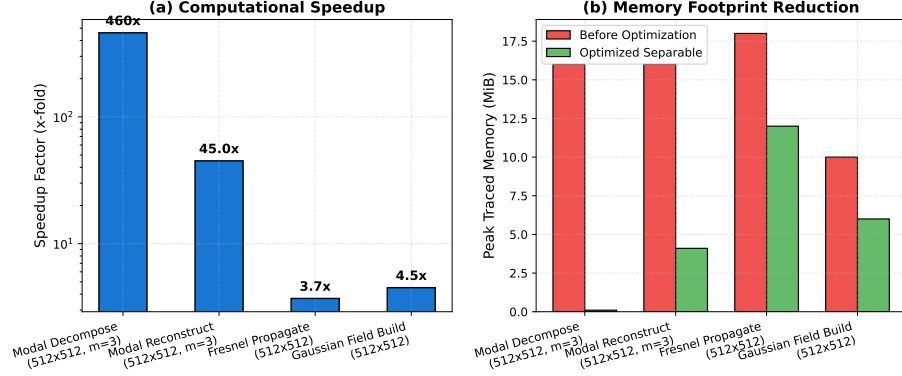


Figure 4: Computational benchmark on 512×512 grids: (a) Speedup factors achieved by vectorized separable kernels; (b) Peak memory allocation comparison showing significant footprint reduction.

1. **Mie Scattering Air Index Default:** When evaluating `miepython` against `PyMieScatt` [7], a consistent 0.2% discrepancy (1.9×10^{-3}) in extinction efficiency Q_{ext} emerged across size parameters $x \in [0.29, 28.6]$. Adjudication against an independent Bohren-Huffman series [8] revealed that `PyMieScatt` defaults to ambient air ($n_{\text{med}} = 1.000273$). Explicitly setting $n_{\text{med}} = 1.0$ restored agreement to 6.4×10^{-14} .
2. **Fresnel Convolution Propagator Anomaly:** In `LightPipes` [9], comparing the convolution propagator (`Fresnel`) against the spectral propagator (`Forward`) and the exact analytical Gaussian beam $w(z) = w_0 \sqrt{1 + (z/z_R)^2}$ revealed that `Fresnel` is consistently 2% $\sim$ 7% wide. Crucially, increasing the grid resolution from 256×256 to 4096×4096 does not reduce the error, demonstrating that the discrepancy originates from the continuous quadratic phase kernel truncation rather than spatial undersampling.

5. Computational Optimization and Separable Modal Expansions

To eliminate runtime overhead, `optcon` adopts boundary-only contract enforcement via the `@checked` decorator. Internal numerical kernels operate directly on standard contiguous NumPy arrays.

By exploiting the spatial separability of Hermite-Gauss modes $u_{mn}(x, y) = u_m(x)u_n(y)$, the 2D modal decomposition of a scalar field $F(x, y)$ on an $N \times N$

grid:

$$C_{mn} = \iint F(x, y) u_m^*(x) u_n^*(y) dx dy, \quad (17)$$

is evaluated as a matrix product:

$$\mathbf{C} = \Delta x^2 (\mathbf{U}^\dagger \mathbf{F} \mathbf{U}^*)^T. \quad (18)$$

As illustrated in Figure 4, this reformulates the decomposition from $\mathcal{O}(M^2 N^2)$ elementwise integrations to two matrix multiplications, reducing execution time on a 512×512 grid from 276 ms to 0.6 ms ($460\times$ acceleration) and cutting peak memory from 16.0 MiB to 0.1 MiB ($160\times$ reduction).

6. Conclusion and Code Availability

`optcon` bridges the gap between high-level machine learning frameworks and rigorous physical semantics in computational optics. By tracking physical dimensions, separating field amplitude from optical power, enforcing invariant operator contracts, and providing rigorous integro-differential solvers for open resonators and nonlinear pulse propagation, it eliminates critical classes of silent failures.

The complete codebase, documentation, tests, and reproducibility benchmarks are open-source and available under the MIT License at: <https://github.com/WhitePepperLambSoup/optcon>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration on Generative AI

During the preparation of this work, the authors utilized large language models (OpenAI ChatGPT/Codex and Google DeepMind Antigravity/Gemini) for code refactoring, docstring formatting, unit test parametrization, and initial draft compilation. All physical formulations, mathematical derivations, software architectures, boundary assertions, numerical benchmarks, and final manuscript texts were independently audited, verified, and validated by the authors. The authors take full responsibility for the contents and findings of this publication.

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